

Architecture Blueprint: High-Scale Multi-Tenant Ebook RAG Platform

Date: April 23, 2026
Objective: Build a scalable, cost-effective platform capable of supporting 4,000+ concurrent tenants, each managing up to 10,000 ebooks, with high-accuracy semantic search and 1-2 paragraph summarization.

1. System Overview

To support a "Future Platform" model, we move beyond simple serverless functions into a **decoupled microservices architecture**. This ensures that a massive upload by one user doesn't degrade the search performance for another.

Component	Technology Recommendation	Role
Frontend	Next.js + Tailwind CSS	Multi-tenant dashboard with drag-and-drop ingestion.
Ingestion Queue	Redis + Celery (on K8s)	Asynchronous processing of 10,000+ file batches.
Vector DB	Milvus (Distributed)	High-speed semantic search with tenant-level isolation.
LLM Orchestration	LangGraph / LlamaIndex	Stateful agents for 2-paragraph synthesis.

2. The Ingestion Pipeline (The "Secret Sauce")

To solve the EPUB issues you experienced with AnythingLLM, this platform uses a multi-stage refinement process:

- **Phase 1: Format Normalization:** Use *EbookLib* to extract clean HTML, then convert to **Markdown** using *Pandoc*. This removes vendor-specific bloat.
- **Phase 2: Global Deduplication:** Before embedding, run a **MinHash LSH** check. If 500 users upload "The Great Gatsby," we store one vector set and reference it via metadata. This reduces costs by up to 80% in a platform model.

- **Phase 3: Semantic Chunking:** Instead of fixed-word counts, use *Sentence Transformers* to break text where the meaning actually shifts.

3. Scalability & Cost Optimization

Autoscaling with KEDA

Using Kubernetes Event-Driven Autoscaling (KEDA), your worker nodes scale to zero when no one is uploading. When a user drops 10,000 books, the system spins up 50+ workers to finish the job in minutes, then shuts them down to save costs.

Hybrid Search Strategy

To provide "Natural" search that actually works, combine:

1. **Dense Retrieval:** Vector search for themes (e.g., "existential dread").
2. **Sparse Retrieval:** BM25 for specific metadata and names (e.g., "Theodore Roosevelt").
3. **Re-ranking:** Use a Cross-Encoder to verify the top 10 results before sending to the LLM.

4. Cost Projections (April 2026)

- **Infrastructure (Base):** ~\$50/mo (Milvus + Redis + Small K8s cluster).
- **Inference:** Pay-per-token using Gemini 1.5 Flash (cheapest high-context model).
- **Storage:** S3-compatible (R2/B2) for raw files (~\$0.015 per GB).

Note: Based on the architecture document provided, this setup supports the 4,000 tenant milestone while maintaining a low entry-level price point.